BEFORE THE
STATE OF NEW YORK
PUBLIC SERVICE COMMISSION

In the Matter of

The Brooklyn union Gas Company d/b/a National Grid NY &
KeySpan Gas East Corporation d/b/a National Grid

Case 19-G-0309 & 19-G-0310

August 2019

Prepared Testimony of:

Staff Information Technology Panel

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Q. Please introduce the members of the Staff Information Technology Panel.
A. The panel members are Jennifer Park, Brian Fisher, Keith Haugen, Luke Quackenbush, Chelsea Kruger
Q. Ms. Park, are you the same Jennifer Park testifying as part of Staff's Gas Supply Panel?
A. Yes, my credentials are provided in that Panel's testimony.
Q. Mr. Fisher, did you provide your educational and professional experience in other testimony in this proceeding?
A. Yes, my professional and educational experience are discussed in the prepared testimony of the Gas Infrastructure and Operations Panel.
Q. Mr. Haugen, in what capacity are you employed by the Department?
A. I am employed by the Department of Public Service as a Utility Analyst 3 – Cyber Security, assigned to the Utility Security Section within the Office of Electric, Gas and Water.
Q. Please provide a summary of your educational and professional experience.
A. I have a B.S. in Information Technology from

Empire State College. I am also certified as a Certified Information Systems Security Professional (CISSP) and a GIAC Systems and Network Auditor (GSNA). Beyond that, I have attended numerous courses and workshops on cyber security. My previous professional work experience consists of 5 years as a computer programmer for Newkirk Products, where I started as a junior programmer and worked my way up to senior programmer. I also became supervisor of my unit, overseeing the work of 6-7 programmers of varying skill levels. For two years following Newkirk, I developed workflow applications for Higher Education Systems Corporation as an IT Specialist 2.

Q. Please describe your current duties with the Utility Security Section of the Office of Reliability and Emergency Preparedness.

A. I joined the Utility Security Section in 2008. My current responsibilities include conducting cyber security vulnerability assessments of critical facilities and corporate IT systems, which are owned and operated by the energy, gas, telecommunications and water utilities.

Q. Have you previously testified before the Commission?

A. Yes. I testified on behalf of the Utility Security Section in Case 16-E-0060 and 16-G-0061, Consolidated Edison Company of New York, Inc.

Q. Mr. Quackenbush, are you the same Luke Quackenbush testifying as part of the Staff Accounting Panel?

A. Yes, my credentials are provided in that Panel's testimony.

Q. Ms. Kruger, are you the same Chelsea Kruger testifying as part of the Staff Consumer Services Panel?

A. Yes, my credentials are provided in that Panel's testimony.

Summary of Testimony

Q. What is the purpose of the Panel's testimony in this proceeding?

A. Our testimony will summarize The Brooklyn Union Gas Company d/b/a National Grid NY's (KEDNY) and KeySpan Gas East Corporation d/b/a National Grid (KEDLI), together the Companies, request for its

new Information Technology (IT) programs and projects, discuss our review process, including the review of Cyber Security and Gas Business Enablement (GBE) projects, recommend a number of adjustments related to the proposed IT projects, and make recommendations to improve the transparency of the Company's IT sanctioning and reporting processes going forward. Our testimony will recommend an overall level of IT Capital Expenditures for KEDNY and KEDLI for the Rate Year, and a level of rent expense to be reflected in the Companies' revenue requirements. It is our position that the Companies should spend at the levels they deem appropriate to provide safe and adequate service.

Q. What adjustments are you recommending to the Companies proposed IT investments?

A. We recommend the following revenue requirements adjustments: (1) an adjustment to address the removal of IT projects; (2) an adjustment to address the removal of the GBE contingency amount; (3) a forecasting adjustment to KEDNY and KEDLI's IT capital budget forecast; and (4)

an adjustment to the KEDNY and KEDLI return on IT capital investments; and (5) a downward-only Service Company Rent IT and GBE Program Net Utility Plant and Depreciation Expense Reconciliation Mechanism. .

Q. Please define the Historic Test Year, Rate Year, and Data Years in these proceedings.

A. The Historic Test Year is the twelve-month period ending December 31, 2018. The Rate Year is the twelve-month period ending March 31, 2021. The Data Years, One, Two, and Three, are subsequent twelve-month periods ending March 31, 2022, March 31, 2023, and March 31, 2024, respectively.

Q. What are KEDNY's and KEDLI's fiscal years (FY)?

A. The Companies' FY runs from April 1 to March 31. They are identified by the calendar year in which the FY ends, such that FY 2021 begins on April 1, 2020, ends on March 31, 2021, and is identical to the Rate Year.

Q. In your testimony, will you refer to, or otherwise rely on, any information obtained during the discovery phase of this proceeding?

A. Yes. We rely on several responses provided by

the Company to information requests, or IRs. These responses are included in Exhibit___(SITP-1) and will be identified using the reference number originally assigned by the Department. For instance, the Department's first IR was identified as "DPS-001."

Q. Is the Panel sponsoring any other exhibits?

A. Yes, we are sponsoring the following additional exhibits: Exhibit___(SITP-2), which includes a summary of, and calculations related to, the Panel's proposed adjustments, and Exhibit___(SITP-3), which shows the GBE adjustment.

The Companies' IT Proposal

Q. What is the role of Information Technology (IT) at National Grid?

A. According to pages 9 and 10 of the pre-filed direct testimony of the Companies' IT Panel, IT serves a wide range of services, from critical gas transmission / distribution support systems to standard office desktop applications, and is vital for the "safe, reliable, and secure physical and commercial operation of KEDNY and

KEDLI's gas distribution business". IT also provides the necessary software applications to serve the customer needs and to efficiently and effectively manage and operate KEDNY and KEDLI.

Q. What does IT provide to the Companies?

A. On page 12 of the pre-filed direct testimony of the Companies' IT Panel, IT "provides, maintains, and manages the computer hardware, software, security (cyber and physical) telecommunications, and other related infrastructure, systems, and services for National Grid's business operations across all National Grid service territories".

Q. Do the Companies develop their own IT projects and investments?

A. No, as most of the IT projects are used by multiple operating companies and subsidiaries of National Grid, IT projects are designed and accounted for by National Grid USA Service Company (Service Company). The associated project costs are allocated to the appropriate operating companies using the allocation factors showed in Exhibit___(ITP-2). The associated project costs for each operating company is in

the form of rent expense.

Q. Describe the total Service Company Rent expense for IT projects and investments to KEDNY and KEDLI in the Historic Test Year and Rate Year.

A. The Service Company Rent expenses associated with IT projects and investments to KEDNY and KEDLI in the Historic Test Year was $27.1 million and $21.3 million, respectively, as shown in Exhibit___(RRP-3CU), Schedule 9. The projected rent expense associated with total IT costs and investments for KEDNY and KEDLI in the Rate Year is $43.7 million and $32.2 million, respectively, as set forth in Exhibit___(RRP-3CU). The Companies claim that the major driver of the increase in rent expense are the forecasted IT capital projects.

Q. What are the types of IT capital programs that the Service Company proposed and why are these investments necessary?

A. There are several types of IT capital programs that the Service Company proposes in this rate case: Technology Modernization; Cyber and Physical Security; Customer Information System (CIS) Replacement; SAP S/4 HANA; Gas Business

Enablement (GBE) Program; Customer Transformation; and other IT capital work streams.

First, Technology Modernization is a program that is aimed to update “National Grid’s IT infrastructure and applications to enable continued reliable and efficient operation”. Some of the major projects in this program include the Application Rationalization project, the Infrastructure Remediation and Lifecycle Refresh project, and the Network Security Infrastructure program.

Second, the Cyber Security Program is aimed to enhance National Grid’s cyber capabilities and “address cyber security threats to the Enterprise, Critical Network Infrastructure (“CNI”), and Operational Technology network systems and environments.” The Physical Security Program is intended to mitigate physical security threats, such as vandalizing property, equipment or facilities, by distributing enhanced security systems across National Grid’s networks and facilities.

Third, the proposed CIS will replace its aging

legacy system in order to have a “fully functional customer system designed to meet customer expectations and address marketplace and regulatory objectives”. A detailed Business Case for CIS is shown in Exhibit___(SSP-5) in the Company’s Shared Service Panel.

Fourth, the SAP S/4 HANA project will replace the Company’s current SAP ECC 6.0, which is a back-office platform that supports various functions, such as Payroll and Human Resources. The main reason for the timing of this decision is because SAP will no longer support the current version National Grid is using after 2025.

Fifth, the GBE Program is “a comprehensive framework of new technology solution and business process changes necessary to strengthen and improve the performance of National Grid’s US gas business.”

Sixth, the Customer Transformation Program would change how National Grid interacts with customers. The program will help “mitigate existing risks to National Grid’s self-service billing, payment, and communications portals”

while improving customer experience through different service options, i.e., web. Seventh and finally, IT will be implementing projects in the following areas: Gas Operations, to help consolidate gas control rooms and upgrades to hardware and operating systems, including current SCADA system; Complex Capital Delivery, to update the IT systems that support the capital delivery service in order to further improve developing and executing the capital plan; Compliance and Mandates, to account for necessary implementation of support capabilities and equipment which are driven by new regulations and safety and reliability requirements; and Corporate IT, to ensure systems are in place to manage and support corporate activities, including Human Resources, assurance, legal, tax, treasury, audit, claims, and other functions.

Q. Are the IT investments divided into spending categories?

A. Yes. As shown in Tables 1 and 2 on page 18 of the Companies' pre-filed Informational Technology Panel testimony, IT Investments are

broken down into the following six categories: Existing Projects, Technology Modernization, Cyber Security, Compliance & Mandates, Gas Business Enablement or GBE, Consumer Information System or CIS, and other initiatives.

The Development of the IT Investment Plan

Q. Describe the Service Company's proposed IT capital spending plan for the Rate Year and FYs 2022 through 2024, which the Companies refer to in their filings as Data Years One, Two and Three.

A. As shown in the Exhibit___(ITP-4CU), the proposed IT spending levels are $243.8 million, $382.8 million, $180.8 million and $299.2 for the Rate Year, FY 22, FY 23, and FY 24, respectively, for the Service Company.

Q. Describe the Companies' Corporate IT budgeting process.

A. As stated in the Companies' response to IR DPS-432, the Companies state that their Corporate IT budgeting process consists of two components: the IT Capital Investment Planning Process and the O&M Planning Process. The IT Capital

Investment Planning Process starts in May, where the budgets are developed, projects and prioritized, and estimates refined for the upcoming fiscal year, which begins the following April. The O&M Planning Process starts in September, where the associated operating expenses are developed by using historical spending trends and estimating the impact of any new projects. In November, the investment plan and O&M budget are implemented in the IT budget process and submitted to the global and U.S. Chief of Information office for approval. After implementation of the investment, different types of monthly meetings are held to discuss actual versus budget variances, updates on investment spending, and validate or review an investment plan that may need to be changed.

Q. Is the Companies' Corporate IT budgeting process appropriate?

A. Yes.

Q. Describe the typical life cycle analysis of an IT project and how the individual project budgets developed.

A. As shown in the response to IR DPS-1003, the

Company explained the five stages of an IT project life cycle: pre start-up, start-up, requirements and design, development and implementation, and close. The pre start-up phase captures the problems and needs to develop scope, context, and cost estimates, which is shown in the Investment Request Paper (IRS). The IRS papers provides a cost estimate of project to have a variance of plus 200 percent or minus 50 percent. Next, the project goes into the start-up phase, where a project manager is assigned to the project to develop a team, work plan, and a partial sanction paper. The partial sanction narrows down the variances of the projects estimates to plus 25 percent and minus 25 percent. After the partial sanction is approved, the project goes into the Development and Implementation phase, where the team determines the technical requirements needed for the project, designs the solution and develops the full sanction paper. The full sanction further reduces the variances of the project estimates to plus 10 percent to minus 10 percent. The sanction paper is submitted to

either the US Sanctioning Committee for projects exceeding $1 million or the IT Sanctioning Committee for project below $1 million. After the sanction is approved, the team builds the solution and tests to see if it operates as required and designed in the development and implementation phase. The solution is implemented, and the transitions begins with necessary support provided. Finally, a closure paper is developed to ensure a clear, efficient, and cost-effective shutdown to the project activity.

Q. Did you, or members of Staff under your supervision, conduct a review of projects in each budget category?

A. Yes, we sampled projects to analyze the status of the projects proposed projects. In the response to IR DPS-487, the Companies provided IRS, partial sanction, and sanction papers for each project depending, on the stage of the project development.

Q. Describe the project sanctioning process.

A. The sanctioning process identifies the appropriate spending levels, by specific

programs or projects. The process is used to seek and obtain approval to spend money on project development. The sanction request may address the full project cost, or a partial sanction may be submitted to request enough funding to advance to the next stage of the project development.

Q. What types of sanctions does the Companies employ for IT capital programs or projects?

A. There are four types of sanctions that the Companies employ: partial sanction, sanction, re-sanction, and closure paper. A partial sanction paper is generally submitted to advance a project when a sanction paper cannot be submitted due to lack of a complete scope and final cost. Unlike the partial sanction paper, the sanction paper is generally prepared for the full scope and cost of the project and is considered the final approval to undertake the project. A re-sanction paper must be filed within the 60 days of notice that the cost of a project is forecasted to exceed the tolerance approved in the sanction paper. A closure paper is prepared at the completion of a project that

details the financial and objective outcomes of the project.

Q. What information is contained in the sanction papers?

A. The sanction papers typically provide cost and project details, as well as potential alternatives and the ramifications of those alternatives, so that the Companies can make informed decisions regarding capital projects, including the risks and benefits to the Companies and their customers. More specifically, the sanction paper includes a summary of the amount being requested for sanctioning, broken down into capital and operating expenditures by year, and a summary of the project, including what is being proposed, what is being replaced, drivers, background, benefits, and any business or customer issues. As some sanctions can be done for multiple projects, a summary of projects is listed. The prior sanctioning history shows each partial or prior sanction before the current sanction paper, along with the sanctioned amount, the next planned sanction, all key milestones, and

the cost estimation tolerance around the sanction requested amount. Each sanction paper categorizes the project as mandatory, policy-driven, justified net present value or other.

Q. What other information do the sanction papers include.

A. Each sanction paper also defines an asset management, risk score, primary risk score driver, complexity level, and process hazard assessment. The resources to complete the project, whether internal or external, availability of the resources, and any potential operational and procurement impact are also noted. Other key issues and potential impact to climate change are also noted. The project alternatives that were considered are listed, along with potential risks faced in project implementation. Any cost assumptions and cost benefit analysis or net present value analysis performed are listed or marked not applicable. The recovery of the project costs and financial impact to the Service Company are defined. If a fully developed sanction is completed, there will be an estimate of expected implementation

operating costs and ongoing run the business expenses. Finally, a list of operating companies that will benefit from – and pay for – the project is included, which a plan for customer outreach, if applicable.

Q. Do all papers in the sanctioning process include all of the information you described?

A. No. Depending on the status of a project's development, it may be in different stages of sanctioning and only preliminary information is included in the documentation.

Q. What information is contained in the IRS papers?

A. IRS papers show the key personnel involved in developing the project, as well as the project category, primary policy driver, description and background, expected benefits, scope, dependencies, and assumptions. Estimated costs by year are listed, as well as costs by delivery phase. A breakdown of the project prioritization and cost by capital, operating, and expected run the business costs is included, along with a score for investment risk and complexity. An estimate of the resources needed to complete the project, the key, known

milestone dates, and benefitting operating companies are also listed.

Q. Do all IRS papers include the information you described above in the sanction process?

A. No. Depending on the status of each individual project, the amount of detail in each IRS paper will vary.

Q. Why is less information available in the IRS papers?

A. IRS papers may contain less information, or more broadly defined information, than full sanction papers because, as described previously, these are used at the earliest stage of project development.

Q. Explain how the Companies estimate the preliminary project costs for IT projects on the IRS papers.

A. As stated in the Companies' response to IR DPS-880, the IT Business Consultants (BC) and Project Managers (PM) develop the preliminary costs, based on their experience with delivering projects of similar size, scope, and complexity. Costs may also be estimated by using labor hours spent on prior projects of similar size, scope,

and complexity, combined with vendor rate cards and internal labor. When applicable, historical documents such as previous statement of work from vendors that show estimated hardware or network costs and sanctioning/closure papers are used.

IT Investment Recovery

Q. How do the Service Companies' IT capital expenditures impact the KEDNY and KEDLI revenue requirements?

A. IT project costs are incurred at the Service Company level. The costs are then allocated to the individual operating companies that use the IT services. KEDNY and KEDLI thus are allocated its proportionate share of IT projects costs for each solution it utilized that was developed or obtained by the Service Company.

Q. What types of IT costs are allocated to KEDNY and KEDLI?

A. The Companies divide their IT program costs into three categories: capital expenditures, operating expenses, and "run the business," or RTB, expenses.

Q. Please describe the capital expenditures category.

A. Capital expenditures represent the costs to buy or create the project that will be included as an asset at the Service Company. Exhibit___(ITP-4CU), shows the forecast capital expenditures, by project, for the Rate Year, as well as FYs 2022, 2023, and 2024.

Q. How do these capital expenditures translate to KEDNY and KEDLI's revenue requirements?

A. Once the project is closed to plant in service on the Service Company's books, the Service Company begins to charge KEDNY and KEDLI for its portion of the amortization expense of the projects, as well as a return on the unamortized project costs. This process is similar to that used for "traditional" gas plant, whereby the Companies incur depreciation expense and also earns a return on the net book value when the plant is included in rate base.

Q. Do the IT assets move to KEDNY or KEDLI's books after being placed in service?

A. No. These assets remain on the Service Company's books after KEDNY and KEDLI begin

using them. The Service Company recovers both the return on and the return of the IT asset investment through Service Company Rent Expense, which is a component of Operations and Maintenance expense, or O&M. Service Company Rent expense is shown in both of KEDNY and KEDLI's Exhibit____(RRP-3), Schedule 9.

Q. How much Service Company Rent expense do KEDNY and KEDLI forecast during the Rate Year?

A. KEDNY and KEDLI forecast $43.7 million and $32.2 million of Rate Year Service Company Rent expense, respectively. Of these amounts, $9.2 million and $10.3 million are facilities, $14.4 million and $8.6 million are for existing gas projects, and $20.1 million and $13.3 million are for new gas IT projects, for KEDNY and KEDLI, respectively.

Q. Please describe the IT operating expenses.

A. As described on page 52 of the Companies' IT Panel's Direct Testimony, IT operating expenses consist of initial project expenses and post-implementation costs incurred with new projects. These costs are spread throughout multiple components of the revenue requirement.

Q. How are operating expenses incurred in the Historic Test Year reflected in the Rate Year revenue requirement?

A. As shown in Exhibit___(ITP-8CU), the Service Company incurred operating expenses of approximately $30.6 million in the historic test year. The Companies expect the level of operating expenses incurred in the historic test year to continue in the Rate Year. After accounting for inflation and allocations to KEDNY and KEDLI, the various components of the Rate Year revenue requirement include approximately $4.271 million and $2.660 million of these expenses for KEDNY and KEDLI, respectively.

Q. What level of operating expenses associated with new IT projects are forecast to be incurred during the Rate Year?

A. As shown in Exhibit___(ITP-7CU), the Service Company expects to incur an additional $33.375 million of operating expenses in the Rate Year associated with new IT projects, not including the CIS project and SAP S4 HANA. After allocating to KEDNY and KEDLI, these forecast

costs result in incremental Rate Year expenses of $3.386 million and $2.109 million for KEDNY and KEDLI, respectively. These expenses are included in the Other Initiatives expense line in O&M, as shown in Exhibit___(RRP-3CU), Schedule 27. Additionally, the calculation is shown in Exhibit___(SITP-2).

Q. Please describe the IT RTB expenses.

A. As explained on page 52 of the Companies' IT Panel Direct Testimony, RTB expenses are costs of running a project once it is in service, inclusive of licensing and maintenance agreements, third party contracts, consultant and contractor costs, and internal labor. Similar to the upfront operating expenses, RTB expenses are included in many areas of the revenue requirement.

Q. How are the RTB expenses incurred in the historic test year reflected in the Rate Year revenue requirement?

A. As shown in Exhibit___(ITP-8CU), the Service Company incurred approximately $213.3 million of RTB expenses in the historic test year. The Companies expect the level of RTB expenses

incurred in the historic rate year to continue into the Rate Year. After accounting for inflation and allocations to KEDNY and KEDLI, the various components of the Rate Year revenue requirement include approximately $29.773 million and $18.544 million of these expenses for KEDNY and KEDLI, respectively.

Q. What level of RTB expenses associated with new IT projects are forecast to be incurred during the Rate Year?

A. As shown in Exhibit___(ITP-7CU), the Service Company expects to incur an additional $28.543 million of RTB expenses in the Rate Year associated with new IT projects, not including the CIS project and SAP S/4 HANA. After allocation to the Companies, these forecast costs result in incremental Rate Year expenses of $2.896 million and $1.803 million for KEDNY and KEDLI, respectively. These expenses are included in the Other Initiatives expense line in O&M, as shown in Exhibit___(RRP-3CU), Schedule 27. Additionally, the calculation is shown in Exhibit___(SITP-2).

Q. Does the Rate Year revenue requirement reflect

upfront operating expenses for CIS?

A. Yes. The Companies included operating expenses of $10.526 million and $4.885 million for KEDNY and KEDLI, respectively, to implement CIS. These expenses are included in the Other Initiatives expense line in O&M, as shown in Exhibit___(RRP-3CU), Schedule 27. Additionally, the calculation is shown in Exhibit___(SITP-2).

Q. Does the Rate Year revenue requirement reflect RTB expenses for CIS?

A. No. As shown on Exhibit___(ITP-7CU), the Companies do not include any Rate Year RTB expenses related to the CIS project because the in-service date does not fall within the Rate Year.

Q. Please summarize KEDNY and KEDLI's Rate Year revenue requirement as it related to IT projects.

A. KEDNY and KEDLI have included approximately $94.598 million and $62.237 million of IT-related project costs in the revenue requirements for their gas businesses, respectively. This is comprised of gas capital-related costs of $43.746 million and $32.236

million for KEDNY and KEDLI, respectively, which are incurred as Service Company Rent expenses; upfront operating expenses of $18.183 million and $9.654 million, and RTB expenses of $32.669 million and $20.347 million for KEDNY and KEDLI, respectively.

Q. How much of this revenue requirement is incremental to what was included in the historic test year and associated with new IT projects?

A. Of the amounts previously provided, approximately $36.932 million and $22.115 million are incremental for KEDNY and KEDLI, respectively. This is comprised of incremental Service Company Rent expense of $20.124 million and $13.318 million, respectively, and operating and RTB expenses of $16.808 million and $8.797 million for KEDNY and KEDLI, respectively.

Staff's Review Process

Q. Describe the process you used to review the Companies proposed IT investments.

A. We took several steps to review the proposed IT investments. First, we held two technical sessions with the Companies to discuss their IT

budgeting process, sanctioning process, proposed IT investment plan and the overall goals and objectives of the Companies' proposed IT investments. Next, we reviewed the Service Company's historic IT capital spending from Fiscal Year 2015 to Fiscal Year 2019 to gauge its ability to complete IT projects. Then, we reviewed the proposed IT projects and associated expenses, through IRS and sanction papers. Finally, we analyzed the variances between the cost estimates of the IT capital projects and the final estimates shown in the closure papers.

Q. Describe the Panel's method of reviewing the proposed projects.

A. Exhibit___(ITP-4CU) shows the IT projects and their associated capital expenditures for the Rate Year and Data Years, totaling 163 IT projects. We focused on reviewing projects that were either starting or continuing in the Rate Year. We reviewed the project using the IRS and sanction papers that were provided in response to IR DPS-487, and found that most of the projects only had IRS papers.

Q. Is the Panel comfortable with most of the

proposed projects only having IRS papers?

A. No. As discussed earlier, IRS papers are generated in the pre start-up phase and show cost estimates that can have a variance of minus 50 percent to plus 200 percent. In addition, the IRS papers do not provide a well-defined scope, therefore, we are not confident that the Companies' can execute the proposed IT investments on time given where the investments currently stand.

Q. What other information did you learn from your review of the Companies' IRS and sanctioning processes?

A. To get a better understanding of how accurate the preliminary project costs shown in the IRS papers or early sanction papers are compared to the final project costs shown in the closure papers, we asked for the variances between the two costs, shown in the Companies' response to IR DPS-880. We found that the variances from the preliminary and final project costs ranged from minus 100 percent to plus 918 percent. We understand that the cost estimates shown on the IRS could range from minus 50 percent to plus

200 percent; however, the variances between the preliminary and final costs show, in actuality, that the Companies' proposed project costs could significantly change and thus significantly impact the Rate Year budget.

Historic Review

Q. What did you observe when you reviewed the historic IT Capital Spending?

A. We made several observations. First, we reviewed the historic IT Capital Spending and found the actual IT capital spending levels in FYs 2015 through 2019 were $74.3 million, $81.4 million, $121.9 million, $94.6 million, and $115.2 million, respectively. Second, we determined that the Service Company underspent its budgets in FY 2015, 2016, 2018, and 2019, but significantly overspent in FY 2017. Lastly, for completed IT projects, we observed significant variances between the preliminary project costs and final costs that were included in the closure papers.

Q. Please explain the variances you have mentioned above.

A. As shown in Exhibit___(SITP-2), which was developed using the Companies' response to IR DPS-432, there were significant variances between the actual and budgeted spending in each of the last five years, FYs 2015 through 2019. In each of those years, the Companies underspent their annual budget by an average of $8.8 million, or 8 percent. The most significant underspend was in FY 2015, when the Companies underspent their $113.8 million budget by $39.6 million, or 35 percent. In FY 2017, however, the Companies' significantly exceeded their budget, spending $122 million, or 84 percent, over their budget of $66.4 million. However, $62.1 million, or 49 percent, of the FY 2017 overspend was incurred in March 2017, which is the last month of the fiscal year.

Q. Explain the reasoning behind the significant overspend in March 2017, the last month of FY 2017.

A. The Companies' stated in response to IR DPS-937 that $54.5 million of the $62.1 million, or 88 percent, of the spend was due to the projects associated for the Technology Improvement

Program (TIP), which was the starting point for Technology Modernization Program. The TIP brought forward several investments that were planned for FY 2018 and beyond, prioritizing the infrastructure, software, and applications that needed to be replaced first.

Q. Does the Panel think the Companies' overspending of their budget in 2017 is an outlier?

A. Yes. The Companies' overspend in FY 2017 was a unique occurrence due to ramping up TIP in one month and we do not expect a significant accelerated spend of capital expenditures like this to happen again.

Q. What is the Panel's opinion of the Companies' proposed IT capital budgets considering their historic IT spending performance?

A. Despite historical IT budgets being significantly lower than the proposed Rate Year Service Company IT budget of $243.8 million, the Companies have consistently underspent their budget. As such, we have concerns that the Companies can deliver on their Rate Year IT budget. While the Service Company has made improvements stemming from the Commission's rate

order in Cases 17-E-0238 and 17-G-0239, these process improvements have not yet resulted in improved performance.

Cyber Security

Q. What do KEDNY and KEDLI propose with respect to cyber security?

A. KEDNY and KEDLI explain that the Service Company proposes to complete 12 projects in the Rate Year, and eight in the Data Years. KEDNY and KEDLI also propose to complete 17 cyber security projects that are expected to be in service after the Historic Test Year, but prior to the Rate Year. There are an additional seven ongoing annual programs that will incur costs throughout the Rate Year and Data Years.

Q. What will these projects address?

A. These projects will address a wide range of cyber security issues, including:

Ensuring adequate security protection to cyber assets supporting critical reliability functions across KEDNY and KEDLI's networks and systems;

Providing security information and event management capabilities to analyze security

alerts in real time, and to collect, store and report on data for compliance and forensic analysis across critical assets and business functions;

Strengthening identity, access and authorization management capabilities to ensure that only authorized personnel can access critical assets and perform only those functions for which they are authorized; and

System modernization cyber security framework utilizing a “defense-in-depth” approach for ensuring a reliable and secure electricity and gas infrastructure and protecting the confidentiality and integrity of sensitive data. Defense in depth, or “layered security” as it is also known, reduces the chance of a successful attack by incorporating redundant security mechanisms. This would require an attacker to circumvent each mechanism to gain access to a digital asset.

Q. What are the proposed costs of these projects?

A. The cyber security projects are forecast to require incremental capital investment of $27.8 million through FY 20, $10.2 million in the Rate

Year, $16.5 million in Data Year 1, $17.7 million in Data Year 2, and $17.3 million in Data Year 3.

Q. Please identify the Companies' five Rate Year Cyber Security proposals that you found especially notable due to cost and complexity.

A. The five Cyber Security proposals are: High Value Discovery & Implementation, CNI Forensic Packet Capture, Enhanced DLP Gateway & Endpoint, CNI Security Enhancements Phase I, and Insider Threat Protection.

Q. Please describe the High Value Discovery & Implementation program.

A. The High Value Asset Discovery & Implementation program, formerly referred to as the IT/OT Discovery and Implementation Phase, is intended to provide KEDNY and KEDLI a holistic view of its company assets across the corporation, including infrastructure, business systems, operational technology and information technology assets. High Value Assets are primary targets for cyber-attacks. Having the ability to effectively identify, control, and monitor High Value Assets is a key component of ensuring

continued security and reliability.

Q. Please describe the CNI Forensic Packet Capture project and how it will enhance the Companies' ability to protect their critical infrastructure?

A. The end goal for the CNI Forensic Packet Capture project is to provide the Companies the capability to capture network packets for forensic analysis to help identify threats laterally across CNI networks. The current solution in place lies on the Internet perimeter gateway, allowing only for analysis of vertical network traffic.

Q. Please describe the CNI Security Enhancements Phase I project and the cyber security concerns it will address?

A. The CNI Security Enhancements Phase I project will address the lack of visibility of cloud services being used through the company and the data that is hosted on those services that can lead to unauthorized access and disclosure of information. As well as lack of visibility and monitoring on those services, the organization has limited ability to enforce security policies

on those services. This project will address these architectural weaknesses.

Q. Please describe the Enhanced DLP Gateway & Endpoint project and the cyber security benefits it will deliver?

A. DLP stands for Data Loss Prevention. The Enhanced DLP Gateway & Endpoint project will deliver new processes and enhance the identity, detection and response capabilities associated with malware attacks by introduction of a DLP gateway at the perimeter. This gateway will inspect network traffic and keep known malware off the network, monitor for malware, and continuously update malware signatures and internet addresses of servers hosting the malware.

Q. Explain the risks of insider threats and how the Insider Threat Protection program will address them.

A. Insider threats pose a significant risk to an organization, having the intimate knowledge of systems and information that if used maliciously can affect operations and put disclosure of confidential information at risk. A high

percentage of breaches are associated with insiders, usually inadvertently by employees. Insider Threat Protection will provide the Companies with the capabilities of continuous monitoring and behavior anomaly detection. The solution will integrate with systems and applications to collect data and analyze users, security events and physical access to identify abnormal behavior associated with data misuse, fraud, or IT sabotage.

Q. What is your overall assessment of the need for the capital and O&M programs described in the Companies' filing?

A. We believe that these investments are needed to address ongoing cyber threats. The implementation of these cyber security countermeasures is essential for establishing a high level of cyber security monitoring and protection. The field of cyber security is one in which the risks, threat actors/vectors, and technologies involved are constantly changing and increasing in complexity at a breakneck pace. As a component of the nation's critical energy infrastructure, there is every reason to

believe that probes and surveillance of the Companies' network and supporting electronic devices will increase in frequency for the foreseeable future, especially as more information is shared at ever-increasing volumes using next-generation mobile devices and cutting-edge technologies, such as cloud-based storage and data analysis tools. Furthermore, the types of information handled on KEDNY and KEDLI's network will inevitably become more critical to their operations in the future and thus, a more tempting target for cyber criminals and hostile nation-states. This ever-escalating cyber threat to business and critical infrastructure information requires that utilities remain constantly vigilant and seek new and innovative ways to bolster their cyber security posture.

Q. Explain how the cyber security programs are developed.

A. Each project in the cyber security program has its own Investment Proposal Summary, benefits case, cost estimate, list of benefitting companies, RTB breakdown, project plan, and

project milestones. The Information Security (IS) Portfolio Delivery Board reviews and approves each project based on that information. A short form Investment Proposal paper for all projects approved by the IS Portfolio Delivery Board is provided to, and formally noted by, the US Sanctioning Committee. Each project will report financial actual spending, forecasts, and budget variances back to the IS Portfolio Delivery Board. The Board will summarize financial data to the program level and manage within the authorized spending cap.

Q. How well has the process you described worked in recent years?

A. Despite the project development process that I just described, actual spending versus budget for cyber security for the previous five rate years was 23% less than forecasted. The Companies have also demonstrated a history of delaying projects due to shifting priorities. For example, the High Value Asset Discovery & Implementation program was delayed due to a review of cyber security strategy in 2018.

Q. What do you recommend?

A. We recommend that KEDNY and KEDLI's cyber security programs be included in the IT project update meetings to ensure timely, consistent progress of the cyber security projects proposed in these proceedings. The IT project update meetings are held quarterly to provide Staff with a progress report on KEDNY and KEDLI's IT projects. We also recommend that the Companies notify appropriate Department Staff as projects get sanctioned or partially sanctioned.

Gas Business Enablement

Q. Please explain the Gas Business Enablement (GBE) Program.

A. The GBE program is an overhaul of National Grid USA's gas business systems that are at or approaching end of life. The overhaul is needed due to the age of the systems in order to support improved safety performance, new capital investment, and evolving regulatory and customer demands.

Q. When did the GBE program commence?

A. The GBE program was proposed by Niagara Mohawk, thoroughly reviewed by Staff, and authorized,

with customer protections, in the Niagara Mohawk Rate Order. The first costs were incurred in FY 2017 for the development of the business case, assessment of processes and applications, and high-level program design. The FY 2017 costs were $20.1 million, which were not included in the approved rates in Case 17-G-0239 and are not proposed for recovery in these cases. The total cost of the GBE program at the Service Company level, $458.1 million - exclusive of the $20.1 million - was extensively vetted in Cases 17-G-0238 and 17-G-0239, as the source of the authorized portion allocated to Niagara Mohawk through rent expense. In addition, there was also a Service Company level contingency of $61.0 million.

Q. Do the Companies propose changes to the GBE program in the instant cases?

A. Yes. The Service Company is transitioning from a multi-vendor integrator to a single system integrator.

Q. Did the Companies explain why?

A. Yes. According to the response to IR DPS-684, the Service Company selected a multi-integrator

approach because it provided the best risk score during evaluation and the Service Company was able to align its vendor with the perceived strengths of program integration. However, the implementation schedule became a challenge and by November 2018, the Service Company determined that the increased complexity of managing dependencies across multiple vendors out-weighed the perceived benefits for which it originally selected a multi-vendor approach. The Service Company worked to simplify the strategy and went through a rebid and recontract for the remaining program scope in order to mitigate increased cost and delayed implementation duration that would result due to the multi-vendor related complications.

Q. Are there any changes to the forecasted budget?

A. The Service Company has sanctioned $57.4 million of the $61.0 million contingency and reflected this amount in the Companies rate request. The increase is related to several changes.

Q. Please describe the changes.

A. The Service Company is planning increased testing and training costs of $12.196 million

related to technical training, Subject Matter Expert testing resources, external testing, and end-user training support. Changes include costs related to greater than anticipated complexity, despite foresight knowing it would be a very difficult and complex integration of the legacy interfaces, of approximately $21.168 million. The final change is an additional $24.025 million for Allowance for Funds Used During Construction (AFUDC). The GBE program initially had $0 for AFUDC because of the "agile" development method the Service Company was using for the proposed program. This approach means working in short-cycles and prioritize functionality and implementing of minimum viable products for portions of the program as they becomes available. In addition, based on the implementation of the program thus far, the Service Company has adjusted various project implementation timelines to ensure a better, smoother rollout.

Q. Does the Panel have any adjustments to the proposed increase of $57.4 million?

A. Yes. At this time, we recommend reflecting $0

of the $57.4 million request in the Companies' rates. The Service Company went through a rigorous development process to forecast the GBE program using two independent consultants; one to develop the costs and roadmap, and the other to ensure cost accuracy and that the program was aligned with industry standards. The Service Company has since run into some trouble with implementation, yet took proactive steps to keep the overall program on a path towards success by moving from a multiple to single system integrator and modifying the roadmap and rollout. However, we recommend, to shield customers from delays and excessive costs, the Service Company work towards a successful on-time and on-budget implementation that provides the forecast customer benefits without seeking recovery of additional expenses from customers during this or subsequent rate proceedings.

Q. How do you recommend reflecting the $57.4 million adjustment?

A. Based on the Companies' responses to IRs DPS-684 and DPS-725, we recommend adjustments to several GBE programs at the Service Company level.

These programs – Asset Management, Asset Management/GIS, and Work Management (Maximo) – and the adjustments to each are shown on Exhibit__(SITP-3). We recommend splitting the $24.1 million adjustment for AFUDC between FY20 and FY21 based on the Companies' response to IR DPS-725, part 3, which stated that only $2.114 million was in FY21, and $21.968 million prior to FY21. The AFUDC adjustments for the previous amounts, and the $12.2 million for testing and training and $21.2 million adjustment for legacy integration complexity, were all based on the forecast FY21 ending balance of each program. The revenue requirement impact of our recommendation on Service Company Rent expense is discussed later in our testimony.

Q. Are you proposing customer protections?

A. Yes, we recommend the same customer protections recommended by the Staff Gas Business Enablement Panel (SGBEP) in Case 17-G-0239 be imposed in the instant cases.

Q. Please describe those protections.

A. The protections include the general protections for all IT investments, which we will recommend

later in our testimony. These include a downward-only reconciliation of net plant and depreciation expense and quarterly reporting. In addition to these general protections, we also recommend additional protections to ensure that the Companies deliver the GBE program benefits, which they identified to support the program, as scheduled and to the full degree envisioned. The additional protections include an overall cap on the amount that can be recovered from customers for GBE, and instituting benchmarks to ensure that the Service Company delivers the incremental benefits of GBE compared to the backbone alternative. Our recommended protections can be found in the Joint Proposal adopted by the Commission in the rate order issued in Cases 17-E-0238 and 17-G-0239. The protections are discussed on pages 44-48 of that Joint Proposal and in Appendix 10 to that Joint Proposal.

Q. What cap do you recommend imposing on the amount KEDNY and KEDLI can recover from ratepayers for GBE?

A. The Companies' response to IR DPS-970 shows the

breakdown of total GBE costs proposed in these cases, $515.4 million, into total capital expenditures, $381.7 million, and total operating expenses, $133.7 million. However, we recommend reducing the proposed capital expenditures by $57.4 million. The result is a cap on the portion of the GBE Project recovered from KEDNY and KEDLI customers of $101.9 million and $47.2 million in total capital expenditures, respectively, and $45.2 million and $20.9 million in total operating expenses, respectively.

Staff Adjustments

Removal of IT Projects

Q. Are you addressing the capital and operating costs associated with any specific IT projects?

A. Yes. We recommend removing the capital expenditures and operating costs associated with the CIS project and the Community Distributed Generation (CDG) Low Income Bill Discount program.

Q. Explain why the Panel recommends removing the CIS project.

A. The reasons for removing the CIS program are discussed in the Staff Consumer Services Panel's testimony.

Q. Explain why the Panel recommends the removal of the CDG Low Income Bill Discount program.

A. As stated in the Companies' responses to IR DPS-620, the wrong allocation code was included on this program, as this program only applies to Niagara Mohawk. The Companies stated in response to IR DPS-882 that the error was identified shortly prior to the corrections and updates and therefore Exhibit__(ITP-4) and the revenue requirement was not updated to remove this program. The Companies agree that an adjustment to the revenue requirement is warranted.

Q. Please explain the Rate Year costs associated with the CIS project.

A. CIS project costs are found in the line items Service Company Rent Expense and Other Initiatives. Capital costs associated with the forecasted project are included the Service Company Rent Expense capital budget, however because the project is not projected to go into

service until October 2022, there is no expense related to the project included in Service Company Rent Expense during the Rate Year. The operating costs associated with CIS, included in Other Initiatives for the Rate Year, are $10.526 million and $4.885 million for KEDNY and KEDLI, respectively.

Q. Please explain the Rate Year costs associated with the CDG Low Income Bill Discount program.

A. The CDG Low Income Bill Discount program project costs are found in the line items Service Company Rent Expense and Other Initiatives. For Service Company Rent expense, the costs for the Rate Year are $0.061 million and $0.032 million for KEDNY and KEDLI, respectively. The operating expense amounts included for this program in Other Initiatives are immaterial.

Q. Do you recommend making an adjustment for these costs?

A. Yes, we are removing the capital and operating costs associated with CIS. For Service Company Rent Expense, because the project is not in service, there are no impacts to the Rate Year, however it does reduce the capital budget. For

Other Initiatives, we recommend removing the operating costs associated with CIS totaling $10.526 million and $4.885 million for KEDNY and KEDLI, respectively. We also recommend removing Service Company Rent expense costs associated with the CDG Low Income Bill Discount program, which amounts to $0.061 million and $0.032 million for KEDNY and KEDLI, respectively.

Removal of GBE Contingency Amount

Q. What is the Panel's adjustment to reflect removal of GBE sanctioned $57.4 million contingency?

A. We are removing capital expenditures of $57.4 million from Service Company Rent expense.

Q. Explain how the removal of $57.4 million was calculated.

A. As stated above, the $57.4 million was allocated to, and removed from, GBE capital costs through the line items 'Asset Management,' 'Asset Management / GIS,' and 'Work Management (Maximo).' Once allocated, and removed, we calculated the adjustment for the return on and return of GBE in the Forecasted Projects Rent

expense model.

Q. What is the impact on Service Company Rent expense in KEDNY and KEDLI's revenue requirements?

A. This adjustment reduces KEDNY and KEDLI Service Company Rent expense by $2.136 million and $1.526 million, respectively.

Forecasting Adjustment

Q. What is the Panel's forecasting adjustment?

A. Our forecasting adjustment reflects a decrease to Rate Year capital expenditures based on a linear regression of the historic actual spending.

Q. What is linear regression?

A. A linear regression is a statistical technique used to model the relationship for a set of data. The results of a linear regression provide an R-squared value, and two coefficients, one being the slope of the line and the other being the y-intercept. Using these two coefficients and the equation of a line (y=mx+b), it is possible to reasonable predict future trends.

Q. What is the R-squared value and what does it represent?

A. The R-squared value is a statistical term that indicates how well the line fits to the data, which ranges from 0.0 to 1.0, 1.0 being a perfect fit. For any R-squared value below 0.9 it was determined that the historical data did not show a high enough correlation and the regression analyses for that group was not utilized in our forecast.

Q. Describe how the Panel used linear regression analysis to develop its forecasted rate year budget.

A. We employed linear regression over FY 2015, 2016, 2018, and 2019. We did not use FY 2017 because we determined the year to be an outlier, as discussed earlier in our testimony.

Q. Describe the results of the linear regressions.

A. As shown in Exhibit___(SITP-2), our analysis of the historic data showed a good statistical fit, an R-squared of 0.932, which shows that the historical data we used was a good fit. We found the forecasted Service Company IT capital expenditures to be $129.4 million.

Q. Why is a forecasting adjustment appropriate in this case?

A. As discussed previously, we are not supportive of the Companies' proposed Rate Year Service Company IT capital expenditures of $243.8 million based on the Companies' historic spending on IT projects and the fact that the proposed IT investments in the Rate Year are primarily based on IRS documents. We acknowledge the fact that the Companies' actual spending compared to the budgeted amount has improved each year in the last five years, however, we recommend the Service Company IT capital expenditures budget be $129.4 million in the Rate Year.

Q. Please explain how the forecasting adjustment impacts the Companies' proposed Rate Year budget.

A. The proposed Rate Year Service Company IT capital expenditures is $243.8 million. We recommend first removing CIS from the total capital budget in the Rate Year, which would result in an adjusted IT capital budget of $173.6 million, or a $70.2 million downward

adjustment. Our recommended Service Company IT capital expenditures budget for the Rate Year is $129.4 million or a 25.5% reduction to our adjusted IT capital budget. In total, with the CIS and forecasting adjustment, there is a total downward adjustment of $114.4 million for the Companies' proposed IT capital expenditures budget.

Q. What is the impact on Service Company Rent expense in KEDNY and KEDLI's revenue requirements?

A. This adjustment reduces revenue requirement for KEDNY and KEDLI Service Company Rent expense by $4.527 million and $2.988 million, respectively. Staff calculated this adjustment by taking the forecast IT projects expense according to the Companies and making the following adjustments: (1) remove the impact of reducing GBE contingency CapEx, (2) remove the impact of reducing the CDG Low Income Program costs, then (3) applying the 25.5 percent reduction to the remaining forecast IT project Rent expense portion.

Q. What is the impact on Other Initiatives expense

in KEDNY and KEDLI's revenue requirements?

A. This adjustment reduces revenue requirement for KEDNY and KEDLI Other Initiatives by $1.600 million and $0.996 million, respectively.

SAP S/4 HANA

Q. Briefly explain the SAP S/4 HANA project.

A. As explained on pages 35 and 36 of the initial testimony of the Companies' IT Panel, the Companies' currently use the SAP ECC 6.0, implemented in 2012, for their back-office platform supporting various essential functions, including Finance, Payroll, Human Resources, and Supply Chain. The Companies' propose to replace the SAP ECC 6.0 with SAP S/4 HANA because SAP will no longer support the SAP ECC 6.0 by 2025. The Companies claim that without the support, they would have the "inability to implement any critical security or compliance patches issued by SAP to address any vulnerabilities discovered, software incompatibility with future products and services offered by SAP, inability to scale in an increasing digital world to take advantage of leading practices, and overall

higher operating costs, decreased performance, and reduced overall reliability."

Q. What is the timeline for this project?

A. On pages 38 and 39 of the initial testimony of the Companies' IT Panel, the Companies stated that they would start their discovery phase in FY 20 and design phase in FY 21.

Q. What is the current status of this project?

A. The Companies have finished the discovery phase of the project. As shown in response to IR DPS-942, the Companies stated that the discovery phase was a 12-week effort that ran from February 2019 to May 2019 and resulted in a "roadmap" for future implementation of the SAP project. The Companies worked with PricewaterhouseCoopers and SAP consultants to map the Companies' current process inventory to a best practice utility standard process; determine the SAP modules that should be included in the solution; define a draft future process scope and structure to improve and optimize current process based on user feedback and lessons learned; and rationalize level of customizations needed based on experience with

current version of SAP. The Companies then developed estimated capital and operating expenditures. The Companies plan to begin the design phase in January 2020 but this may change as this phase has not yet been approved or sanctioned.

Q. Briefly describe what the Companies propose for project costs associated with the SAP S/4 HANA project.

A. As shown on page 7, Exhibit___(ITP-4CU), and Exhibit___(ITP-8CU) in the Companies' corrections and updates filing, the Companies propose a Service Company capital and operating expenditure of $48.6 million $30.6 million, respectively. The Companies also propose to defer for future recovery the differences between the costs recovered in rates and the Companies' actual costs, as explained on page 40 of the initial testimony of the Companies' IT Panel.

Q. How did the Companies develop the capital and operating expenditures for the SAP S/4 HANA project?

A. According to the Companies' response to IR DPS-

942, the Companies developed the estimated capital and operating expenditures utilizing a PricewaterhouseCoopers model. The model utilizes a detailed bottom up resource plan for labor for all parts of the project and adds the estimated non-labor costs associated with the project.

Q. Does the Panel agree with this project?

A. Yes. We agree with the need for the project and the proposed operating expenditures needed for the design phase of the project. However, due to the design phase not being sanctioned yet and uncertainty as to when the design phase will actually begin, the Companies should adjust the capital expenditures for the project for the Rate Year to work within our forecasted budget of $129.4 million.

Q. Does the Panel agree with the Companies' proposal to defer the differences between the costs recovered in rates and the Companies' actual costs for future recovery from customers?

A. No. We understand that the Companies' proposed costs are preliminary estimates and may significantly vary; however, we believe

customers should not be required to pay if the Companies' overspend the budget they developed in the discovery phase. Instead, as we will describe later in our testimony, we recommend a downward only mechanism for all IT projects.

Service Company Asset Recovery Charge

Q. As discussed earlier, the IT project assets remain on the Service Company's books after KEDNY and KEDLI begin using them. The Service Company recovers the return on the IT asset investment from KEDNY and KEDLI, and the other National Grid operating companies, through Service Company Rent Expense. What rate of return did the Companies request to apply to the return on of IT capital costs in the Rate Year?

A. The Companies proposed to use a pre-tax weighted average cost of capital of 8.69 percent in their original filings, which is based on a Return on Equity, or ROE, of 9.65 percent with a capital structure comprised of 50 percent common equity and 50 percent long-term debt. This is shown on pages 16 and 17 of the direct testimony of the Companies' witness Jonathan Cohen. The

Companies updated this amount during Corrections and Updates, filed on July 3, 2019. In the update filing, the Companies proposes to use an updated pre-tax weighted average cost of capital of 8.52 percent.

Q. Do you agree with Mr. Cohen's proposal to use the Service Company rate of return, which includes a 50 percent common equity ratio?

A. No. We understand that the Staff Finance Panel is recommending for KEDNY and KEDLI a common equity ratio of 48 percent and a return on equity of 8.20 percent. Accordingly, we recommend that the common equity ratio and cost rates for common equity and long-term debt recommended by the Staff Finance Panel also should be used in the development of revenue requirement for Service Company Rent expense. This would result in a pre-tax weighted average cost of capital of 7.58 percent and 7.54 percent for KEDNY and KEDLI, respectively. We recommend this rate be applied to assets at the Service Company level so as to avoid requiring customers to pay a higher return on these assets solely because the Companies have chosen to have a

Service Company provide the assets.

Q. What is your adjustment for this reduction in the use of the Staff Finance Panel rate of return?

A. This adjustment reduces KEDNY and KEDLI Service Company Rent expense by $3.553 million and $2.805 million, respectively.

Adjustments Summary

Q. Please summarize your revenue requirement adjustments related to IT projects.

A. For KEDNY and KEDLI, respectively, our revenue requirement adjustments decrease Service Company Rent expense by $10.322 million and $7.355 million; and IT Operating and RTB expenses, which are included in Other Initiatives expense, by $12.126 million and $5.881 million.

Downward Only Reconciliation

Q. Is the Panel concerned that the Service Company will under-spend its Rate Year IT budget?

A. Yes. As previously discussed, the Service Company's historical data shows that there have been significant variances between the capital

budget and actual expenditures. As discussed earlier, the Service Company is planning a substantial increase in IT spending. However, the Service Company has not provided enough support to show that it can ramp-up its spending to complete all of the proposed IT investments. For these reasons, there is a significant risk that the Service Company will again fail to execute its spending plan fully and timely, thereby forcing KEDNY's and KEDLI's customers to pay rates based on a level of new plant that is not actually deployed.

Q. Does the Panel's forecasted budgeting adjustment address this concern?

A. Partially. Our forecasted budget adjustment, as well as the adjustment to remove CIS, reduces the allowed Service Company capital IT spending to $129.4 million in the Rate Year. However, despite this reduction to the Companies' request of $243.8 million, it still exceeds the IT capital spend in prior years by a significant amount. FY 18 and 19 had IT capital spends of $94.6 million and $115.2 million, respectively. Although FY 17 reported an IT capital spend of

$122.0 million, we have concerns about the data for that period, as we previously discussed.

Q. What do you recommend to address this concern?

A. Similar to what the Commission approved in the Niagara Mohawk Rate Order, we recommend a Service Company Rents IT and GBE Program Net Utility Plant and Depreciation Expense Reconciliation Mechanism to protect customers from paying delivery rates higher than necessary to support the IT assets actually implemented.

Q. Briefly describe the proposed Service Company Rents IT and GBE program Net Utility Plant and Depreciation Expense Reconciliation Mechanism.

A. Our recommended mechanism would require the Companies to reconcile its respective actual IT and GBE program average net utility plant and depreciation expense revenue requirements to the forecast revenue requirements. The IT and GBE program average net utility plant and depreciation expense revenue requirement would be calculated by applying the Companies' authorized pre-tax weighted average cost of capital in the Rate Year to the IT and GBE program average net utility plant balance and

adding the depreciation expense to the product. At the end of the Rate Year, if the actual IT and GBE program average net utility plant and depreciation expense revenue requirement is less than the target average net utility plant and depreciation expense revenue requirement, the Companies would defer the revenue requirement impact for the benefit of customers.

Q. Should this mechanism be symmetrical?

A. No. The mechanism should be a one-way, downward only true-up. Therefore, if actual Service Company Rent expense exceeds the approved Rate Year allowance, a regulatory liability would not be established for the Companies to recover from customers at a later date. The calculations for the effects of this mechanism for the Rate Year should be filed with the Secretary on or before July 31st after the close of the Rate Year.

Q. Why does the Panel recommend that the mechanism be a one-way, downward-only true-up mechanism?

A. Budgeting and spending are activities wholly within the Companies' control. Improving their performance in these areas also is within the Companies' control. A symmetrical true-up will

not provide an incentive for the Companies to improve their budgeting and spending processes. Customers have no control over the Companies' level and pace of spending, yet they bear the risk that the Companies' historic challenges in spending to projected levels will continue and will be reflected in rates. The true-up mechanism, therefore, should reconcile only underspending to allocate these risks equitably between the Companies and customers.

Process Improvements

Q. Do you have any recommendations for process improvements?

A. Yes. We recommend to: (1) continue the process improvements that were required in the Niagara Mohawk Rate Order; (2) make improvements to the Service Company's IRS and sanction papers; and (3) implement Key Performance Indicators (KPIs) for projects.

Q. Please explain the process improvements required in the Niagara Mohawk Rate Order.

A. The improvements addressed reporting and processes for IT. For reporting requirements,

Niagara Mohawk was required to make quarterly and annual filings. Prior to each of its rate years, Niagara Mohawk is required to file its IT Capital Investment Plan for that upcoming rate year, including the overall IT investment plan for that year, the process for developing the plan, and identifying the proposed IT projects and their estimated costs. Niagara Mohawk is also required to file quarterly variance reports within 60 days of the end of each rate year quarter to address the status of the Service Company IT programs/projects and budgets. The variance reports include (i) a narrative explaining overall IT investment plan approach; (ii) a description of defined program categories that form the spending rationale for IT projects; (ii) detail on budgets and actual sending for each of the program categories, and for the top ten projects by budget within the program categories; (iv) explanations of variances between budgets and actual spending; (v) identification of allocations to Niagara Mohawk; and (vi) a report on budget expectations by program category and top ten projects based

on allocations to Niagara Mohawk. The variance report in the last quarter would also include (i) an explanation of any cost or timeline differences exceeding ten percent for projects completed; and the latest, partial, full, re-sanction or closure paper for any projects exceeding $1 million that were authorized during the previous rate year.

Q. Are there additional requirements?

A. Yes. Niagara Mohawk is required to meet with Staff on a semi-annual basis to (i) discuss proposed projects for the next six months; (ii) provide an update on the status of in-flight projects; (iii) discuss proposed projects for the next six months; and (iv) review quarterly filings. The Commission required that, in the first meeting, Niagara Mohawk work with Staff to develop a mutually agreeable sanction paper enhancements needed to justify the IT investments and allow for adequate review. The enhancements would include minimum cost alternative analyses, as appropriate, based on the type of IT investment (e.g., regular upgrades versus build of new function), and the

consensus enhancements were to be implemented no later than the end of that first rate year. Niagara Mohawk and Staff would also determine if additional or different information would be useful for the quarterly variance reports.

Q. Does the Panel recommend the same reporting and process requirements, which were approved in the Niagara Mohawk Rate Order, for KEDNY and KEDLI?

A. Yes, we recommend the Companies implement the same reporting requirements and meet with Staff on a semi-annual basis to discuss the topics mentioned above. We realize that the reporting and meeting can be combined for all New York operating companies for efficiency purposes. We also recommend additional reporting and process requirements.

Q. Please explain your recommendation for additional reporting and process requirements.

A. We recommend the following additional requirements to be included in the variance reports: charts showing the governance structure for each of the top ten projects; identification of major implementation risks and the plans to manage & mitigate them; and an explanation of

the root cause(s) of said variances & plans to eliminate the root causes(s). For the semi-annual meetings, we recommend that the following also be discussed: updates on the status of any attendant shut-down operations that are supposed to produce savings; review strategic-level risks and issues related to the implementation of in-flight projects; and discuss any salient changes to governance, contracted resources, or staffing.

Q. Should these reporting requirements continue beyond the Rate Year?

A. Yes. We believe it is important for the Commission to monitor the Companies' capital investment plans on an ongoing basis. Informational reports at regular intervals are critical to maintain oversight of the IT investment plans. These recommendations are consistent with existing reporting requirements for the Companies' gas operations and for Niagara Mohawk. They are critical to establishing a foundation for Staff to conduct comprehensive, project-specific examinations of IT projects in future rate proceedings,

comparable to Staff's current examination of gas capital plans.

Q. Please explain your recommendations regarding information provided in the IRS and sanction papers.

A. We recommend that the Companies more fully support its cost estimates and work collaboratively with Staff to show that such estimates are reasonable. We also recommend that the Service Company IT sanction paper or IRS document should state if the solution chosen was the minimum-cost alternative. If the Companies chose a higher cost, or enhanced, program, the sanction paper should present an analysis that compares the benefits and costs associated with the project life cycle. The analysis should also further explain how the results support the decision to pursue the alternative solution.

Q. Please explain your recommendations for the KPIs.

A. We recommend the Companies develop KPIs to track and quantify benefits stated in Sanction papers and compare and report on the actual benefits

versus what was originally proposed.

Q. Does this conclude your testimony at this time?

A. Yes.